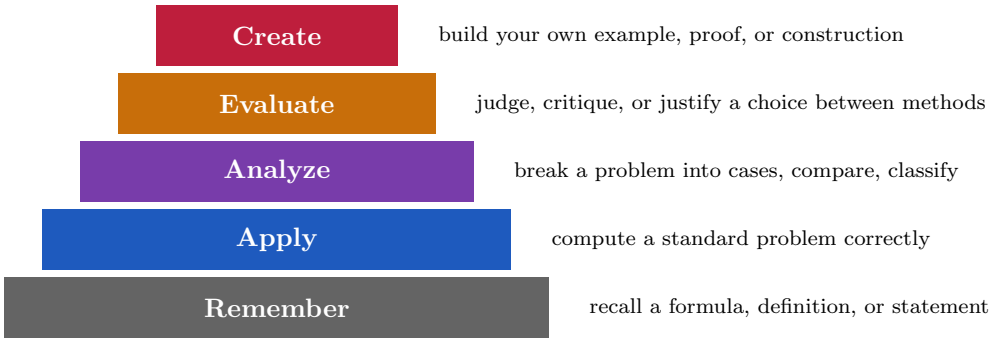
BAMAT101

Practice Problem Bank

Modules 1–5 — Across Bloom's Taxonomy

How to use this document. Every topic of the syllabus gets at least five questions spanning several levels of Bloom's Taxonomy — not just "compute this", but also recalling, explaining, comparing, critiquing and constructing your own examples. Exam questions increasingly test whether you understand *why* a method works, so the Analyze, Evaluate and Create questions deserve real time.
Attempt every question before checking the Answer Key at the end. For Analyze/Evaluate/Create questions, write your reasoning out in full sentences — that is exactly the skill an exam rewards.

Suggested routine. First pass: Remember and Apply questions for all topics. Second pass: Analyze, Evaluate and Create. Re-attempt anything you could not do unaided a few days later.



Syllabus Coverage Map

Module	Topics (question label prefix)
Module 1 Functions of Several Variables	Functions of Two Variables and Partial Derivatives (F) Total Differential (TD) Jacobian and Its Properties (J) Taylor's Expansion for Two Variables (TE) Maxima and Minima of Functions of Two Variables (MM) Constrained Maxima and Minima: Lagrange's Multiplier Method (LM)
Module 2 Multiple Integrals	Double Integrals: Basic Evaluation and Change of Order (DI) Double Integrals in Polar Coordinates (DP) Triple Integrals: Cylindrical and Spherical Coordinates (TI)
Module 3 Vector Differentiation and Integration	Gradient (G) Directional Derivative (D) Divergence (Dv) Curl (C) Scalar and Vector Potential (P) Line Integrals (L) Green's Theorem (Gr) Surfaces and Surface Integrals (S) Stokes' Theorem (St) Gauss Divergence Theorem (Gs)
Module 4 Ordinary Differential Equations	Complementary Function (Homogeneous Case) (O) Undetermined Coefficients and Resonance (PI) Method of Variation of Parameters (VP) Cauchy–Euler Equation (CE) Cauchy–Legendre Equation (CL) Applications of Linear Second-Order ODEs (AP)
Module 5 Partial Differential Equations	Formation of Partial Differential Equations (FP) Solutions of the Type $F(p, q) = 0$ (PQ) Solutions of the Type $F(z, p, q) = 0$ (ZP) Solutions of the Type $F(x, p) = G(y, q)$ (XP) Clairaut's Form and the Singular Solution (CK) Lagrange's Linear Equation $Pp + Qq = R$ (LG) Linear First-Order PDEs by Separation of Variables (SV)

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Answer Key **19**

Module 1 — Functions of Several Variables

1 Functions of Two Variables and Partial Derivatives

F1. [Remember] Define $\partial f/\partial x$ for a function $f(x,y)$ as a limit, and state the condition under which the mixed partials satisfy $f_{xy} = f_{yx}$.

F2. [Apply] Find f_x, f_y and f_{xy} for $f(x,y) = x^3y^2 + \sin(xy)$, and check that f_{yx} gives the same result.

F3. [Apply] Show that $u = \ln(x^2+y^2)$ satisfies Laplace's equation $u_{xx}+u_{yy} = 0$ for $(x,y) \neq (0,0)$.

F4. [Apply] If $z = x^2y$ with $x = t^2$ and $y = t^3$, find dz/dt using the chain rule $\frac{dz}{dt} = z_x\frac{dx}{dt} + z_y\frac{dy}{dt}$, and confirm by substituting first.

F5. [Analyze] Let $f(x,y) = \dfrac{xy}{x^2+y^2}$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$. Show that $f_x(0,0)$ and $f_y(0,0)$ both exist, yet f is not continuous at the origin.

F6. [Evaluate] A student asserts that $\dfrac{\partial^2 f}{\partial x\,\partial y} = \dfrac{\partial^2 f}{\partial y\,\partial x}$ for *every* function. Evaluate this claim.

F7. [Create] Construct a non-constant function $u(x,y)$ satisfying Laplace's equation $u_{xx}+u_{yy} = 0$, and verify it.

2 Total Differential

TD1. [Remember] State the total differential dz of $z = f(x,y)$ and say what it approximates.

TD2. [Apply] For $z = x^2y - 3xy^2$ find dz. Evaluate it at $(1,2)$ with $dx = 0.02$, $dy = -0.01$ and compare with the exact change Δz.

TD3. [Apply] Use differentials to approximate $\sqrt{(4.02)^2 + (2.97)^2}$.

TD4. [Apply] A cylinder's radius and height are measured as $r = 5$ cm and $h = 12$ cm with possible errors of 0.1 cm and 0.2 cm. Estimate the maximum error in its volume $V = \pi r^2 h$, and the relative error.

TD5. [Analyze] For $z = x^2 + y^2$ at $(1,1)$ with $dx = dy = h$, compute Δz and dz. Show that $\Delta z - dz$ is of second order in h, and explain why this matters.

TD6. [Evaluate] A student estimates the change in $f = x^2 + y^2$ from $(1,1)$ to $(3,3)$ using dz and reports $dz = 8$, calling it "exact since the differential is the exact change". Evaluate.

TD7. [Create] Construct a function $z = f(x,y)$ for which $dz = \Delta z$ exactly for all dx, dy, and explain why it works.

3 Jacobian and Its Properties

J1. [Remember] Define the Jacobian $\dfrac{\partial(u,v)}{\partial(x,y)}$, and state its reciprocal property and chain property.

J2. [Apply] Find $\dfrac{\partial(x,y)}{\partial(r,\theta)}$ for $x = r\cos\theta,\ y = r\sin\theta$.

J3. [Apply] If $u = x^2 - y^2$ and $v = 2xy$, find $\dfrac{\partial(u,v)}{\partial(x,y)}$ and hence $\dfrac{\partial(x,y)}{\partial(u,v)}$.

J4. [Apply] Compute the Jacobian $\dfrac{\partial(x,y,z)}{\partial(\rho,\varphi,\theta)}$ for spherical coordinates $x = \rho\sin\varphi\cos\theta,\ y = \rho\sin\varphi\sin\theta,\ z = \rho\cos\varphi$.

J5. [Analyze] Show that $u = x + y$ and $v = x^2 + 2xy + y^2$ are functionally dependent, and find the relation between them.

J6. [Evaluate] A student argues: "for $u = x^3,\ v = y$ the Jacobian is $3x^2$, which is 0 at $x = 0$, so u and v are functionally dependent." Evaluate.

J7. [Create] Construct a transformation $x = x(u,v),\ y = y(u,v)$ with constant Jacobian $\dfrac{\partial(x,y)}{\partial(u,v)} = 6$, and use it to find the area of the image of the unit square $0 \le u, v \le 1$.

4 Taylor's Expansion for Two Variables

TE1. [Remember] Write Taylor's expansion of $f(x,y)$ about (a,b) up to terms of the third degree.

TE2. [Apply] Expand $f(x,y) = e^x\cos y$ about $(0,0)$ up to second-degree terms.

TE3. [Apply] Expand $f(x,y) = x^2y + 3y - 2$ in powers of $(x-1)$ and $(y+2)$.

TE4. [Apply] Use a second-degree Taylor expansion of $f(x,y) = x^y$ about $(1,1)$ to approximate $(1.1)^{1.02}$.

TE5. [Analyze] Find the second-degree Taylor polynomial of $f(x,y) = \ln(1+x+y)$ about $(0,0)$ in two ways: (i) from partial derivatives, (ii) by substituting $t = x+y$ into the series for $\ln(1+t)$. Compare the two routes.

TE6. [Evaluate] A student uses the linear (first-degree) Taylor polynomial of $f = x^2 + y^2$ about $(0,0)$ to estimate $f(0.1, 0.1)$, gets 0, and concludes Taylor's method is useless. Evaluate.

TE7. [Create] Construct a function $f(x,y)$ whose second-degree Taylor polynomial about $(0,0)$ is $1 + 2x - y + x^2 + 3xy$.

5 Maxima and Minima of Functions of Two Variables

MM1. [Remember] State the second-derivative test for a critical point of $f(x,y)$, using $r = f_{xx},\ s = f_{xy},\ t = f_{yy}$.

MM2. [Apply] Find and classify the critical points of $f(x,y) = x^3 + y^3 - 3xy$.

MM3. [Apply] An open-topped rectangular box must hold 32 m^3. Find the dimensions that minimise the surface area.

MM4. [Analyze] For $f = x^4 + y^4 - 2x^2 + 4xy - 2y^2$, find all critical points and classify them. The

test is inconclusive at one of them — decide its nature by another method.

MM5. [Evaluate] A student claims $f = x^2 - y^2$ has a minimum at the origin "because along the x-axis $f = x^2 \geq 0$". Evaluate.

MM6. [Create] Construct a function with exactly one critical point, which is a local minimum at $(2, -1)$.

6 Constrained Maxima and Minima: Lagrange's Multiplier Method

LM1. [Remember] State Lagrange's condition for extremising $f(x, y, z)$ subject to the constraint $g(x, y, z) = 0$.

LM2. [Apply] Find the point on the plane $x + 2y + 3z = 14$ closest to the origin, and the minimum distance.

LM3. [Apply] Find the maximum and minimum of $f = x + 2y$ on the circle $x^2 + y^2 = 5$.

LM4. [Apply] Find the dimensions of the rectangular box of greatest volume, with faces parallel to the coordinate planes, inscribed in the sphere $x^2 + y^2 + z^2 = 12$.

LM5. [Analyze] Explain geometrically why ∇f must be parallel to ∇g at a constrained extremum. Illustrate with the circle-and-line example $f = x + 2y$ on $x^2 + y^2 = 5$.

LM6. [Evaluate] To extremise $f = x^2 + y^2$ subject to $xy = 1$, a student finds the points $(1, 1)$ and $(-1, -1)$ with $f = 2$ and declares it the *maximum*, since it is the only candidate. Evaluate.

LM7. [Create] Construct a Lagrange problem — an objective f and a linear constraint — whose constrained maximum is at $(2, 3)$.

Module 2 — Multiple Integrals

7 Double Integrals: Basic Evaluation and Change of Order

DI1. [Remember] Explain why the order of integration does not affect the value over a rectangular region, but the limits must be adjusted carefully for a non-rectangular one.

DI2. [Apply] Evaluate $\int_0^1 \int_0^x (x+y)\,dy\,dx$.

DI3. [Apply] Change the order of integration in $\int_0^1 \int_x^{\sqrt{x}} f(x,y)\,dy\,dx$ and write the reversed-order integral.

DI4. [Analyze] Evaluate $\int_0^1 \int_x^1 e^{y^2}\,dy\,dx$ by first changing the order of integration (the original order has no elementary antiderivative). Show the sketch-based reasoning for the new limits.

DI5. [Evaluate] A student reverses $\int_0^1 \int_0^{1-x} f\,dy\,dx$ to $\int_0^1 \int_0^{1-y} f\,dx\,dy$ without sketching, calling the swap "automatic". Evaluate whether this swap is correct, and the general risk of skipping the sketch.

DI6. [Create] Construct a region bounded by a parabola and a line of your choice, and set up the double integral of a function f over it in both orders, $dy\,dx$ and $dx\,dy$.

8 Double Integrals in Polar Coordinates

DP1. [Remember] State the polar substitution formulas and explain why $dA = r\,dr\,d\theta$ rather than just $dr\,d\theta$.

DP2. [Apply] Evaluate $\iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dA$ by converting to polar coordinates.

DP3. [Apply] Evaluate $\iint_R xy\,dA$, where R is the quarter disk $x^2+y^2 \le 4$ in the first quadrant.

DP4. [Analyze] Explain why evaluating $\iint_R \sqrt{x^2+y^2}\,dA$ over a disk directly in Cartesian coordinates is much harder than in polar coordinates.

DP5. [Evaluate] For the annulus between circles of radius 1 and 2, a student sets the limits $r : 0 \to 2,\ \theta : 0 \to 2\pi$. Evaluate the error and give the corrected limits.

DP6. [Create] Construct a double integral over the circular sector with $30° \le \theta \le 90°$ and $0 \le r \le 3$, and write the polar limits.

9 Triple Integrals: Cylindrical and Spherical Coordinates

TI1. [Remember] State the dV expressions for cylindrical and spherical coordinates, and one geometric clue that suggests each.

TI2. [Apply] Use cylindrical coordinates to find the volume bounded by the cylinder $x^2+y^2=4$

and the planes $z = 0,\ z = 5$.

TI3. **[Apply]** Use spherical coordinates to evaluate $\iiint_V z\,dV$, where V is the upper half of the ball $x^2 + y^2 + z^2 \le 1$.

TI4. **[Analyze]** A solid lies between the cone $z = \sqrt{x^2 + y^2}$ (below) and the sphere $x^2+y^2+z^2 = 2$ (above). Find the correct φ-limits in spherical coordinates, explaining how the cone fixes a value of φ.

TI5. **[Evaluate]** A student sets up a triple integral over a solid cylinder using spherical coordinates and gets very messy limits. Evaluate this choice and explain why cylindrical coordinates fit better.

TI6. **[Create]** Describe (in words) a solid for which spherical coordinates are clearly the best choice, and justify using its symmetry.

Module 3 — Vector Differentiation and Integration

10 Gradient

G1. [Remember] State the formula for ∇f in Cartesian coordinates, and its geometric relationship to the level surface through a point.

G2. [Apply] Find ∇f for $f(x, y, z) = x^3y - 2yz^2 + xz$ at $(1, 2, -1)$.

G3. [Apply] Find the tangent plane to $x^2 + 2y^2 + 3z^2 = 12$ at $(1, 2, -1)$.

G4. [Analyze] For $f(x, y) = x^2 - y^2$, describe how the level curves change shape as c passes through 0, and explain what happens to ∇f at the origin.

G5. [Evaluate] A student claims: "wherever $\nabla f = \mathbf{0}$, f has a local maximum or minimum." Evaluate, giving a counterexample if false.

G6. [Create] Construct a scalar function $f(x, y, z)$ whose gradient at $(1, 1, 1)$ equals $(2, 3, -1)$.

11 Directional Derivative

D1. [Remember] Write the formula for the directional derivative of f along a unit vector $\hat{\mathbf{a}}$, and explain why $\hat{\mathbf{a}}$ must be a unit vector.

D2. [Apply] Find the directional derivative of $f = xy^2 + yz$ at $(2, -1, 3)$ in the direction of $(3, 4, 0)$.

D3. [Apply] Find the maximum rate of change of $f = x^2y^3$ at $(2, 1)$ and the direction in which it occurs.

D4. [Analyze] At $(2, 1)$ on $f = x^2 + y^2$, without fully computing, decide which of the directions $(1, 0)$ and $(0, 1)$ gives the larger directional derivative, using the angle each makes with ∇f.

D5. [Evaluate] Explain why the directional derivative perpendicular to ∇f is always zero, then verify for $f = x^2 + y^2$ at $(1, 1)$.

D6. [Create] Construct $f(x, y)$ and a point P such that the maximum directional derivative at P is exactly 5.

12 Divergence

Dv1. [Remember] State the formula for $\operatorname{div} \mathbf{F}$ and the physical meaning of positive and negative divergence.

Dv2. [Apply] Compute $\operatorname{div} \mathbf{F}$ for $\mathbf{F} = x^2y\,\mathbf{i} - 2xz\,\mathbf{j} + yz^2\,\mathbf{k}$.

Dv3. [Apply] Find λ such that $\mathbf{F} = (2x + 3y)\mathbf{i} + (y - z)\mathbf{j} + (\lambda z)\mathbf{k}$ is solenoidal.

Dv4. [Analyze] For $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$, compute $\operatorname{div} \mathbf{F}$ and explain why it is the same constant at every point, not only near the origin where the arrows seem to "start".

Dv5. [Evaluate] A field has $\operatorname{div} \mathbf{F} = 0$ everywhere except at the origin, where it is undefined. Can

it be called solenoidal away from the origin, and what goes wrong if the Divergence Theorem is applied to a surface enclosing the origin?

Dv6. [Create] Construct a vector field that is a uniform sink with $\operatorname{div}\mathbf{F} = -6$ at every point.

13 Curl

C1. [Remember] State the determinant formula for $\operatorname{curl}\mathbf{F}$ and the physical meaning of $\operatorname{curl}\mathbf{F} = \mathbf{0}$.

C2. [Apply] Compute $\operatorname{curl}\mathbf{F}$ at $(1,1,1)$ for $\mathbf{F} = xz\,\mathbf{i} + xyz\,\mathbf{j} - y^2\mathbf{k}$.

C3. [Apply] Determine whether $\mathbf{F} = (2xy + z^2)\mathbf{i} + (x^2 + 2z)\mathbf{j} + (2xz + 2y)\mathbf{k}$ is irrotational.

C4. [Analyze] Explain, using the determinant formula, why $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ has zero curl everywhere.

C5. [Evaluate] A student computes $\operatorname{curl}\mathbf{F} = 2\mathbf{k}$ for $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$ but argues "the field lines are just circles, so no particle is really spinning". Evaluate.

C6. [Create] Give a vector field that is solenoidal but not irrotational.

14 Scalar and Vector Potential

P1. [Remember] State the condition for a scalar potential φ to exist for $\mathbf{F}$, and the relationship between them.

P2. [Apply] Verify that $\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$ is irrotational and find its scalar potential.

P3. [Apply] Verify that $\mathbf{F} = (y + z)\mathbf{i} + (x + z)\mathbf{j} + (x + y)\mathbf{k}$ is irrotational and find its scalar potential.

P4. [Analyze] Explain why a scalar potential, when it exists, is unique only up to an additive constant.

P5. [Evaluate] $\mathbf{F}$ is solenoidal and a student tries to find φ with $\mathbf{F} = \nabla\varphi$, but gets stuck. Is this expected, and what should be sought instead?

P6. [Create] Construct a nonzero vector field that is both solenoidal and irrotational.

15 Line Integrals

L1. [Remember] State the definition of $\int_C \mathbf{F}\cdot d\mathbf{r}$ and what it represents physically.

L2. [Apply] Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ for $\mathbf{F} = (x + y)\mathbf{i} + (y - x)\mathbf{j}$ along the straight line from $(0,0)$ to $(2,2)$.

L3. [Apply] Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ for $\mathbf{F} = y\mathbf{i} + x\mathbf{j}$ along $y = x^3$ from $(0,0)$ to $(1,1)$.

L4. [Analyze] For $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$, evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ from $(1,0)$ to $(0,1)$ along (a) the straight segment and (b) the quarter circle $x^2 + y^2 = 1$. Compare, and explain in terms of curl.

L5. [Evaluate] Student A computes $\oint_C \mathbf{F} \cdot d\mathbf{r} = +8$ going counterclockwise; Student B gets -8 going clockwise. Can both be right?

L6. [Create] Construct a vector field and two paths between the same two points along which the line integral differs.

16 Green's Theorem

Gr1. [Remember] State Green's Theorem and the orientation of C it requires.

Gr2. [Apply] Use Green's Theorem to evaluate $\oint_C (3x^2y\,dx + x^3\,dy)$ around the boundary of $0 \le x \le 1,\ 0 \le y \le 2$, counterclockwise.

Gr3. [Apply] Use Green's Theorem to evaluate $\oint_C \left(xy^2\,dx + (x^2y + 2x)\,dy\right)$ around the triangle $(0,0), (1,0), (1,1)$, counterclockwise.

Gr4. [Analyze] $\mathbf{F} = \dfrac{-y}{x^2+y^2}\mathbf{i} + \dfrac{x}{x^2+y^2}\mathbf{j}$ is undefined at the origin. Why can't Green's Theorem be applied directly to a curve enclosing the origin, and what is the standard fix?

Gr5. [Evaluate] A closed boundary consists of 6 line segments with a moderately complex integrand. Which is likely faster: parametrising each segment, or Green's Theorem?

Gr6. [Create] Construct M, N such that $\oint_C (M\,dx + N\,dy) = \text{Area}(R)$ for every region R.

17 Surfaces and Surface Integrals

S1. [Remember] Write dS for a surface $z = g(x,y)$, and say what the square-root factor accounts for.

S2. [Apply] Find the flux of $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + 2z\mathbf{k}$ upward through the part of $x + y + z = 1$ in the first octant.

S3. [Apply] Find the surface area of the paraboloid $z = x^2 + y^2$ below $z = 4$.

S4. [Analyze] Explain why reversing the normal flips the sign of a flux integral but not its magnitude, and give a scenario where the sign matters.

S5. [Evaluate] A full sphere is not the graph of one function $z = g(x,y)$. Why can't $dS = \sqrt{1 + g_x^2 + g_y^2}\,dA$ be used directly, and what is the fix?

S6. [Create] Construct a flat surface and a nonzero constant field whose flux through it is exactly zero.

18 Stokes' Theorem

St1. [Remember] State Stokes' Theorem and the orientation convention linking C and $\hat{\mathbf{n}}$.

St2. [Apply] Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$, with C the triangle $(1,0,0), (0,1,0), (0,0,1)$ oriented outward.

St3. [Apply] Use Stokes' Theorem for $\mathbf{F} = y^2\mathbf{i} + xy\mathbf{j} + xz\mathbf{k}$ around the square $0 \le x, y \le 2$ in $z = 0$, counterclockwise from $+z$.

St4. [Analyze] A flat disk and a hemispherical bowl share the same boundary circle C. Why must $\iint (\operatorname{curl} \mathbf{F}) \cdot \hat{\mathbf{n}}\, dS$ be equal over both?

St5. [Evaluate] $\mathbf{F}$ is irrotational on a simply connected region. What does that imply about $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for every closed curve there?

St6. [Create] Construct a closed curve and a field for which a flat disk is clearly the easiest capping surface, and explain.

19 Gauss Divergence Theorem

Gs1. [Remember] State the Gauss Divergence Theorem and what kind of surface it requires.

Gs2. [Apply] Verify the Divergence Theorem for $\mathbf{F} = 2x\mathbf{i} + y\mathbf{j} + 3z\mathbf{k}$ over the cube $[0, 1]^3$ by computing both sides.

Gs3. [Apply] Use the Divergence Theorem to evaluate $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS$ for $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$ over the sphere $x^2 + y^2 + z^2 = a^2$.

Gs4. [Analyze] Why is the flux of $\mathbf{F} = \hat{\mathbf{r}}/r^2$ through *any* closed surface enclosing the origin the same constant?

Gs5. [Evaluate] A student applies the Divergence Theorem directly to an *open* hemispherical surface (curved part only). Is that valid?

Gs6. [Create] Construct a field whose flux through every closed surface is zero.

Module 4 — Ordinary Differential Equations

20 Complementary Function (Homogeneous Case)

O1. [Remember] State the three root cases of the auxiliary equation of $ay'' + by' + cy = 0$ and the form of the complementary function for each.

O2. [Apply] Solve $y'' - 7y' + 12y = 0$.

O3. [Apply] Solve $y'' + 6y' + 9y = 0$.

O4. [Apply] Solve $y'' - 4y' + 13y = 0$.

O5. [Analyze] Using the discriminant, classify each as oscillatory, purely exponential, or repeated-root: (a) $m^2 + 4m + 4 = 0$, (b) $m^2 + 4 = 0$, (c) $m^2 + 5m + 6 = 0$.

O6. [Evaluate] A student leaves the solution of $y'' + 2y' + 5y = 0$ as $C_1 e^{(-1+2i)x} + C_2 e^{(-1-2i)x}$. Is this acceptable, and why is the real form preferred?

21 Undetermined Coefficients and Resonance

PI1. [Remember] State the trial form for the particular integral when $R(x)$ is (a) an exponential, (b) a degree-n polynomial, (c) a sine or cosine, and when it must be modified.

PI2. [Apply] Find the particular integral of $y'' - 3y' + 2y = e^{3x}$.

PI3. [Apply] Solve $y'' + 4y = 8x^2$ completely.

PI4. [Apply] (Resonance) Solve $y'' - 4y' + 4y = e^{2x}$, noting the naive guess collides with a repeated root.

PI5. [Analyze] For $y'' - y' - 2y = 3e^{2x} + \sin x$ (roots $2, -1$), decide for each term whether resonance occurs and give the trial forms.

PI6. [Evaluate] A student solving $y'' + y = \sin x$ with the guess $A\cos x + B\sin x$ gets $0 = \sin x$. What went wrong, and what is the correct guess?

PI7. [Create] Construct a second-order ODE with real distinct roots and initial conditions so the solution is a sum of two exponentials with positive coefficients; solve it.

22 Method of Variation of Parameters

VP1. [Remember] State the variation-of-parameters formula for $y'' + Py' + Qy = R$ and when it is preferred over undetermined coefficients.

VP2. [Apply] Solve $y'' + y = \sec x$.

VP3. [Apply] Solve $y'' - 2y' + y = \dfrac{e^x}{x}$.

VP4. [Apply] Find a particular integral of $y'' + 4y = \tan 2x$.

VP5. [Analyze] Why does undetermined coefficients fail for $y'' + y = \tan x$, while variation of parameters works?

VP6. [Evaluate] A student finds $y_p = -xe^x + xe^x \ln x$ for $y'' - 2y' + y = e^x/x$ and calls it wrong because $-xe^x$ lies in the CF. Evaluate.

VP7. [Create] Construct a constant-coefficient ODE that *requires* variation of parameters, and set up (without solving) the two integrals.

23 Cauchy–Euler Equation

CE1. [Remember] State the Cauchy–Euler form, the substitution that solves it, and the resulting auxiliary equation.

CE2. [Apply] Solve $x^2y'' - 2xy' - 4y = 0$.

CE3. [Apply] Solve $x^2y'' + 3xy' + y = 0$.

CE4. [Apply] Solve $x^2y'' + xy' + y = 0$.

CE5. [Apply] Solve $x^2y'' - 3xy' + 4y = x^2$.

CE6. [Analyze] Show that $x = e^t$ gives $xy' = \dfrac{dy}{dt}$ and $x^2y'' = \dfrac{d^2y}{dt^2} - \dfrac{dy}{dt}$, and say why this makes the coefficients constant.

CE7. [Evaluate] A student solves $x^2y'' + xy' - 4y = 0$ by assuming $y = e^{mx}$. Evaluate, and give the right approach.

CE8. [Create] Construct a Cauchy–Euler equation whose general solution is $y = C_1x^3 + C_2x^{-2}$.

24 Cauchy–Legendre Equation

CL1. [Remember] State the Cauchy–Legendre form and its substitution; how does it relate to Cauchy–Euler?

CL2. [Apply] Solve $(1 + x)^2y'' + (1 + x)y' + y = 0$.

CL3. [Apply] Solve $(2x + 3)^2y'' - 2(2x + 3)y' - 12y = 6x$.

CL4. [Apply] Solve $(3x + 2)^2y'' + 3(3x + 2)y' - 36y = 0$.

CL5. [Evaluate] A student tries $y = x^m$ on $(x + 1)^2y'' + (x + 1)y' - y = 0$. Evaluate and give the correct trial.

CL6. [Create] Construct a Cauchy–Legendre equation in $(2x + 1)$ whose general solution is $y = C_1(2x + 1)^2 + C_2(2x + 1)^{-1}$.

25 Applications of Linear Second-Order ODEs

AP1. [Remember] State the ODEs for a damped spring–mass system and a series LCR circuit, and the three damping regimes.

AP2. [Apply] A 0.5 kg mass on a spring with $k = 8$ N/m is pulled 0.3 m and released from rest (no damping). Find $x(t)$, the amplitude and the period.

AP3. [Apply] Solve $x'' + 6x' + 9x = 0,\ x(0) = 1,\ x'(0) = 0$ and classify the damping.

AP4. [Apply] A series LCR circuit has $L = 1$ H, $R = 4\ \Omega$, $C = \frac{1}{5}$ F, a constant EMF $E = 10$ V, and starts with $q(0) = 0,\ i(0) = 0$. Find $q(t)$.

AP5. [Analyze] Solve $x'' + 4x = \cos 2t$ and describe what happens physically.

AP6. [Evaluate] A student says that because $x'' + 2x' + 5x = 0$ has complex roots, the system "oscillates forever". Evaluate.

AP7. [Create] Choose the damping coefficient c for $m = 1,\ k = 4$ so the system is critically damped, and give the general solution.

Module 5 — Partial Differential Equations

26 Formation of Partial Differential Equations

FP1. [Remember] State the two methods of forming a PDE (eliminating constants and eliminating arbitrary functions), and the order of the PDE each tends to produce. Notation: $p = z_x,\ q = z_y$.

FP2. [Apply] Form the PDE by eliminating a, b from $z = (x + a)(y + b)$.

FP3. [Apply] Form the PDE by eliminating a, b from $z = ax + by + a^2 + b^2$.

FP4. [Apply] Form the PDE by eliminating the arbitrary function f from $z = f(x^2 + y^2)$.

FP5. [Apply] Form the PDE by eliminating ϕ from $\phi(x + y + z,\ x^2 + y^2 + z^2) = 0$.

FP6. [Analyze] Show that eliminating the arbitrary functions f, g from $z = f(x + y) + g(x - y)$ gives $\dfrac{\partial^2 z}{\partial x^2} = \dfrac{\partial^2 z}{\partial y^2}$, and explain why the order is two here.

FP7. [Evaluate] Eliminating a, b from $z = ax + by + ab$ gives $z = px + qy + pq$. A student claims every solution of this PDE has the form $z = ax + by + ab$. Evaluate.

FP8. [Create] Construct a first-order PDE satisfied by every function $z = f(y/x)$.

27 Solutions of the Type $F(p, q) = 0$

PQ1. [Remember] Describe the method of solving $F(p, q) = 0$.

PQ2. [Apply] Find the complete integral of $p + q = pq$.

PQ3. [Apply] Find the complete integral of $\sqrt{p} + \sqrt{q} = 1$.

PQ4. [Apply] Find the complete integral of $q = p^2$.

PQ5. [Analyze] The assumed solution $z = ax + by + c$ has three constants, yet a complete integral should have two. Explain.

PQ6. [Evaluate] A student claims the only solutions of $p^2 + q^2 = 1$ are the planes $z = x\cos\alpha + y\sin\alpha + c$. Evaluate.

PQ7. [Create] Construct a PDE $F(p, q) = 0$ whose complete integral is $z = ax + \dfrac{1}{a}y + c$.

28 Solutions of the Type $F(z, p, q) = 0$

ZP1. [Remember] Describe the method for $F(z, p, q) = 0$ (no explicit x, y).

ZP2. [Apply] Solve $p(1 + q) = qz$.

ZP3. [Apply] Solve $z^2(p^2 + q^2 + 1) = 1$.

ZP4. [Apply] Solve $pq = z$.

ZP5. [Analyze] Why does the substitution $z = f(x + ay)$ work for $F(z, p, q) = 0$, and why does it need the absence of explicit x, y?

ZP6. [Evaluate] A student applies $u = x + ay$ to $pq = xz$. Evaluate.

ZP7. [Create] Construct a PDE $F(z, p, q) = 0$ and solve it by $u = x + ay$ to get an exponential complete integral.

29 Solutions of the Type $F(x, p) = G(y, q)$

XP1. [Remember] Describe the method for $F(x, p) = G(y, q)$.

XP2. [Apply] Solve $p^2 + q^2 = x + y$.

XP3. [Apply] Solve $p^2 y(1 + x^2) = qx^2$.

XP4. [Apply] Solve $pe^y = qe^x$.

XP5. [Analyze] Why can both sides of $F(x, p) = G(y, q)$ be set equal to the *same* constant?

XP6. [Evaluate] A student says $p + q = x + y$ cannot be solved by this method "because x and y are mixed". Evaluate.

XP7. [Create] Construct a PDE of this type whose complete integral is $z = (x + a)^2 + (y - a)^2 + b$.

30 Clairaut's Form and the Singular Solution

CK1. [Remember] State Clairaut's form of a first-order PDE and its complete integral.

CK2. [Apply] Find the complete and singular solutions of $z = px + qy + p^2 + q^2$.

CK3. [Apply] Find the complete and singular solutions of $z = px + qy + \sqrt{1 + p^2 + q^2}$.

CK4. [Apply] Reduce $(p + q)(z - xp - yq) = 1$ to Clairaut's form and write its complete integral.

CK5. [Analyze] The complete integral of $z = px + qy + p^2 + q^2$ is a two-parameter family of planes. What does the singular solution $4z = -(x^2 + y^2)$ represent geometrically?

CK6. [Evaluate] A student says $z = -xy$ solves $z = px + qy + pq$ "by taking $a = -y,\ b = -x$ in the complete integral". Evaluate.

CK7. [Create] Construct a Clairaut equation whose singular solution is the paraboloid $z = x^2 + y^2$.

31 Lagrange's Linear Equation $Pp + Qq = R$

LG1. [Remember] State Lagrange's equation, its auxiliary (subsidiary) equations, and the form of the general solution.

LG2. [Apply] Solve $xp + yq = z$.

LG3. [Apply] Solve $yz\,p + zx\,q = xy$.

LG4. [Apply] Solve $(y-z)p + (z-x)q = x-y$.

LG5. [Apply] Solve $(y+z)p + (z+x)q = x+y$.

LG6. [Analyze] Interpret $\dfrac{dx}{P} = \dfrac{dy}{Q} = \dfrac{dz}{R}$ geometrically and explain why $\phi(u,v) = 0$ represents a solution surface.

LG7. [Evaluate] Having found $u = y/x$, $v = z/x$ for $xp + yq = z$, a student writes the general solution as $u + v = c$. Evaluate.

LG8. [Create] Construct a Lagrange equation whose general solution is $z = f(x^2 + y^2)$.

32 Linear First-Order PDEs by Separation of Variables

SV1. [Remember] Describe the method of separation of variables for a first-order linear PDE.

SV2. [Apply] Solve $\dfrac{\partial u}{\partial x} = 2\dfrac{\partial u}{\partial t} + u$, given $u(x,0) = 6e^{-3x}$.

SV3. [Apply] Solve $3\dfrac{\partial u}{\partial x} + 2\dfrac{\partial u}{\partial y} = 0$, given $u(x,0) = 4e^{-x}$.

SV4. [Apply] Solve $4u_x + u_y = 3u$, given $u(0,y) = e^{-5y}$.

SV5. [Analyze] In the earlier problem $u_x = 2u_t + u$ the condition fixed $k = -3$. If instead $u(x,0) = 6e^{-3x} + 4e^{-5x}$, how does the solution change?

SV6. [Evaluate] From $\dfrac{X'}{X} = 2\dfrac{T'}{T} + 1$, a student argues both sides must be 0 "since X depends only on x and T only on t". Evaluate.

SV7. [Create] Construct a first-order linear PDE and a condition whose separated solution is $u = 5e^{2x+3y}$.

Answer Key

Remember-level questions are answered with the core fact itself; Apply-level questions with the final result (and key steps); Analyze/Evaluate/Create questions with a model reasoning path — your wording can differ as long as the reasoning is equivalent.

Module 1 — Functions of Several Variables

Functions of Two Variables and Partial Derivatives

F1 $\dfrac{\partial f}{\partial x} = \lim\limits_{h\to 0} \dfrac{f(x+h,y)-f(x,y)}{h}$, with y held fixed. The mixed partials are equal whenever both are continuous near the point (Clairaut–Schwarz theorem).

F2 $f_x = 3x^2y^2 + y\cos(xy)$, $f_y = 2x^3y + x\cos(xy)$, and $f_{xy} = f_{yx} = 6x^2y + \cos(xy) - xy\sin(xy)$.

F3 $u_x = \dfrac{2x}{x^2+y^2}$, so $u_{xx} = \dfrac{2(y^2-x^2)}{(x^2+y^2)^2}$. By symmetry $u_{yy} = \dfrac{2(x^2-y^2)}{(x^2+y^2)^2}$, and the two cancel: $u_{xx} + u_{yy} = 0$.

F4 $z_x = 2xy = 2t^5$, $dx/dt = 2t$, $z_y = x^2 = t^4$, $dy/dt = 3t^2$, so $dz/dt = 4t^6 + 3t^6 = 7t^6$. Check: $z = t^7$, so $dz/dt = 7t^6$. ✓

F5 On both axes $f = 0$, so $f_x(0,0) = \lim_{h\to 0} \frac{f(h,0)-0}{h} = 0$ and likewise $f_y(0,0) = 0$. But along the line $y = x$, $f = \frac{1}{2}$ for every $x \neq 0$, so the limit at the origin along that line is $\frac{1}{2} \neq f(0,0)$: f is discontinuous. Partial derivatives only probe the two axis directions, so their existence does not guarantee continuity.

F6 False in general; equality needs the mixed partials to be continuous. Counterexample: $f = \dfrac{xy(x^2-y^2)}{x^2+y^2}$, $f(0,0) = 0$. Here $f_x(0,y) = -y$ and $f_y(x,0) = x$, so (writing $f_{xy} = \partial_y(f_x)$) $f_{xy}(0,0) = -1$ while $f_{yx}(0,0) = +1$.

F7 E.g. $u = x^2 - y^2$: $u_{xx} = 2$, $u_{yy} = -2$, sum 0. Other valid choices: xy, $e^x\cos y$, $\ln(x^2+y^2)$.

Total Differential

TD1 $dz = f_x\,dx + f_y\,dy$. It approximates the actual change $\Delta z = f(x+dx,\, y+dy) - f(x,y)$ for small dx, dy — it is the change measured along the tangent plane.

TD2 $dz = (2xy - 3y^2)\,dx + (x^2 - 6xy)\,dy$. At $(1,2)$: $z_x = -8$, $z_y = -11$, so $dz = -0.16 + 0.11 = -0.05$. Exact: $\Delta z = -10.0475 - (-10) = -0.0475$.

TD3 Take $f = \sqrt{x^2+y^2}$ at $(4,3)$: $f = 5$, $f_x = 4/5$, $f_y = 3/5$, with $dx = 0.02$, $dy = -0.03$. Then $df = 0.016 - 0.018 = -0.002$, so the value is ≈ 4.998 (exact: 4.99813).

TD4 $dV = 2\pi rh\,dr + \pi r^2\,dh = 2\pi(5)(12)(0.1) + \pi(25)(0.2) = 12\pi + 5\pi = 17\pi \approx 53.4$ cm^3. Relative error $= \dfrac{17\pi}{300\pi} = \dfrac{17}{300} \approx 5.7\%$.

TD5 $\Delta z = (1+h)^2 + (1+h)^2 - 2 = 4h + 2h^2$ and $dz = 2h + 2h = 4h$, so $\Delta z - dz = 2h^2$.

The discrepancy shrinks like h^2, much faster than h itself, which is why dz is a good linear approximation for small steps.

TD6 Incorrect. $dz = 2(1)(2) + 2(1)(2) = 8$, but the true change is $\Delta z = 18 - 2 = 16$ — a 50% error. The differential is only a *linear approximation*, reliable for small dx, dy; a step of 2 is far too large.

TD7 Any affine function, e.g. $z = 3x - 2y + 5$: $\Delta z = 3\,dx - 2\,dy = dz$. It works because f_x, f_y are constants and there are no second-order terms to create a discrepancy.

Jacobian and Its Properties

J1 $\dfrac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = u_x v_y - u_y v_x$. Reciprocal: $\dfrac{\partial(u,v)}{\partial(x,y)} \cdot \dfrac{\partial(x,y)}{\partial(u,v)} = 1$. Chain: $\dfrac{\partial(u,v)}{\partial(r,s)} = \dfrac{\partial(u,v)}{\partial(x,y)} \cdot \dfrac{\partial(x,y)}{\partial(r,s)}$.

J2 $\begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r$.

J3 $\begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4x^2 + 4y^2$, so $\dfrac{\partial(x,y)}{\partial(u,v)} = \dfrac{1}{4(x^2+y^2)}$.

J4 $\rho^2 \sin\varphi$ — exactly the factor in $dV = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta$ (Module 2).

J5 $\dfrac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 1 & 1 \\ 2x+2y & 2x+2y \end{vmatrix} = 0$ identically, so they are dependent. Indeed $v = (x+y)^2 = u^2$.

J6 Incorrect. Functional dependence needs $J \equiv 0$ on a whole region. Here $J = 0$ only on the line $x = 0$; u and v are independent. What fails at $x = 0$ is local invertibility ($x = u^{1/3}$ is not differentiable at 0).

J7 E.g. $x = 2u,\ y = 3v$: $J = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6$. The image is a 2×3 rectangle of area $|J| \cdot 1 = 6$.

Taylor's Expansion for Two Variables

TE1 With $h = x - a,\ k = y - b$ and all derivatives evaluated at (a,b): $f(a+h, b+k) = f + hf_x + kf_y + \frac{1}{2!}\big(h^2 f_{xx} + 2hk\, f_{xy} + k^2 f_{yy}\big) + \frac{1}{3!}\big(h^3 f_{xxx} + 3h^2 k\, f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\big) + \cdots$

TE2 At $(0,0)$: $f = 1,\ f_x = 1,\ f_y = 0,\ f_{xx} = 1,\ f_{xy} = 0,\ f_{yy} = -1$. So $f \approx 1 + x + \frac{1}{2}(x^2 - y^2)$.

TE3 With $h = x - 1,\ k = y + 2$ (the expansion terminates, so it is exact): $f = -10 - 4h + 4k - 2h^2 + 2hk + h^2 k$.

TE4 At $(1,1)$: $f = 1,\ f_x = 1,\ f_y = 0,\ f_{xx} = 0,\ f_{xy} = 1,\ f_{yy} = 0$. So $f \approx 1 + h + hk$ with $h = 0.1,\ k = 0.02$: $1 + 0.1 + 0.002 = 1.102$ (exact: 1.10210).

TE5 (i) $f(0,0) = 0$, $f_x = f_y = 1$, $f_{xx} = f_{xy} = f_{yy} = -1$, giving $(x+y) - \frac{1}{2}(x^2 + 2xy + y^2)$. (ii)

$t - \frac{t^2}{2}$ with $t = x + y$ gives the same. Substitution is much faster whenever f depends on a single combination of x and y.

TE6 At the origin $f_x = f_y = 0$, so the linear polynomial is 0 — the true value is 0.02. The method is not at fault; the first nonzero terms are of second degree. Including them: $\frac{1}{2}(2h^2 + 2k^2) = h^2 + k^2 = 0.02$, which is exact.

TE7 The polynomial itself works, as does any function with $f(0,0) = 1,\ f_x = 2,\ f_y = -1,\ f_{xx} = 2,\ f_{xy} = 3,\ f_{yy} = 0$ at the origin, e.g. $f = 1 + 2x - y + x^2 + 3xy + x^3$.

Maxima and Minima of Functions of Two Variables

MM1 At a point where $f_x = f_y = 0$: if $rt - s^2 > 0$ and $r < 0$, local maximum; if $rt - s^2 > 0$ and $r > 0$, local minimum; if $rt - s^2 < 0$, saddle point; if $rt - s^2 = 0$, the test is inconclusive.

MM2 $f_x = 3x^2 - 3y = 0,\ f_y = 3y^2 - 3x = 0$ give $(0,0)$ and $(1,1)$. With $r = 6x,\ s = -3,\ t = 6y$: at $(0,0)$, $rt - s^2 = -9 < 0$, a saddle; at $(1,1)$, $rt - s^2 = 27 > 0$ and $r = 6 > 0$, a local minimum with $f = -1$.

MM3 Let base $x \times y$, height $z = 32/(xy)$. $S = xy + 2z(x + y) = xy + \dfrac{64}{y} + \dfrac{64}{x}$. Setting $S_x = y - \dfrac{64}{x^2} = 0$ and $S_y = x - \dfrac{64}{y^2} = 0$ gives $x = y = 4$, $z = 2$, and $S_{\min} = 48$ m^2.

MM4 Critical points: $(0,0),\ (\sqrt{2}, -\sqrt{2}),\ (-\sqrt{2}, \sqrt{2})$. At the latter two, $r = t = 20$, $s = 4$, so $rt - s^2 = 384 > 0$ with $r > 0$: local minima, $f = -8$. At $(0,0)$: $r = t = -4,\ s = 4$, so $rt - s^2 = 0$ (inconclusive). Along $y = x$: $f = 2x^4 \geq 0$; along $y = -x$: $f = 2x^4 - 8x^2 < 0$ for small $x \neq 0$. f takes both signs near the origin, so $(0,0)$ is a saddle.

MM5 Incorrect. Checking one direction is not enough: along the y-axis $f = -y^2 \leq 0$. Formally $r = 2,\ s = 0,\ t = -2$ gives $rt - s^2 = -4 < 0$, so the origin is a saddle point.

MM6 $f = (x-2)^2 + (y+1)^2$: $f_x = f_y = 0$ only at $(2,-1)$, where $r = t = 2,\ s = 0$, so $rt - s^2 = 4 > 0$ and $r > 0$.

Constrained Maxima and Minima: Lagrange's Multiplier Method

LM1 Solve $\nabla f = \lambda \nabla g$ together with $g = 0$ (equivalently, find stationary points of $f - \lambda g$). The scalar λ is the Lagrange multiplier; geometrically the gradients of f and g are parallel at the extremum.

LM2 Minimise $x^2 + y^2 + z^2$: $2x = \lambda,\ 2y = 2\lambda,\ 2z = 3\lambda$, so $x = \lambda/2,\ y = \lambda,\ z = 3\lambda/2$. The constraint gives $7\lambda = 14$, $\lambda = 2$: point $(1,2,3)$, distance $\sqrt{14}$.

LM3 $(1,2) = \lambda(2x, 2y)$ gives $y = 2x$; then $5x^2 = 5$, $x = \pm 1$. Maximum $f = 5$ at $(1,2)$; minimum $f = -5$ at $(-1,-2)$.

LM4 Maximise $V = 8xyz$ ($x, y, z > 0$ the half-edges). $yz = 2\lambda x,\ xz = 2\lambda y,\ xy = 2\lambda z$ force $x^2 = y^2 = z^2$, so $x = y = z = 2$. The box is a cube of edge 4, $V_{\max} = 64$.

LM5 The constraint curve $g = 0$ is a path along which we may move. If the level curve of

f crossed the constraint curve (not tangent), we could slide along the constraint to a higher or lower f-value, so we would not be at an extremum. At an extremum the level curve of f just touches the constraint curve, so their normals $\nabla f, \nabla g$ are parallel. In the example, the lines $x+2y=c$ touch the circle exactly at $(1,2)$ ($c=5$) and $(-1,-2)$ ($c=-5$).

LM6 Incorrect. Lagrange's method only gives candidates; their nature must be checked. Here $f \to \infty$ along the hyperbola (e.g. $x=t,\ y=1/t$ gives $f=t^2+t^{-2}\to\infty$), so $f=2$ is the *minimum*; there is no maximum.

LM7 E.g. maximise $f=xy$ subject to $3x+2y=12$. Then $(y,x)=\lambda(3,2)$ gives $y=3x/2$, hence $x=2,\ y=3,\ f_{\max}=6$ (check: on the line $f=x(12-3x)/2$, maximised at $x=2$).

Module 2 — Multiple Integrals

Double Integrals: Basic Evaluation and Change of Order

DI1 Over a rectangle, both variables have constant limits independent of each other, so the order can be swapped freely. Over a non-rectangular region, the inner limits are boundary curves expressed in terms of the outer variable; swapping the order means re-reading the boundary from a sketch, not merely relabelling.

DI2 Inner: $\int_0^x (x+y)\,dy = x^2+\frac{x^2}{2}=\frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2}\,dx=\frac{1}{2}$.

DI3 The region lies between $y=x$ and $y=\sqrt{x}$ for $0\le x\le 1$. For fixed $y\in[0,1]$, x runs from y^2 (the curve $y=\sqrt{x}$) to y (the line $y=x$): $\displaystyle\int_0^1\int_{y^2}^{y} f(x,y)\,dx\,dy$.

DI4 The region is the triangle $0\le x\le y\le 1$. Reversed: y from 0 to 1, x from 0 to y. Then $\int_0^1\int_0^y e^{y^2}\,dx\,dy=\int_0^1 ye^{y^2}\,dy=\dfrac{e-1}{2}$.

DI5 It happens to be correct: the triangle $x,y\ge 0,\ x+y\le 1$ is symmetric under $x\leftrightarrow y$. But that is luck — for most regions (e.g. bounded by a parabola) the swap is not symmetric, and skipping the sketch gives wrong limits.

DI6 E.g. region between $y=x^2$ and $y=x$: $\displaystyle\int_0^1\int_{x^2}^{x} f\,dy\,dx=\int_0^1\int_{y}^{\sqrt{y}} f\,dx\,dy$. (Any valid pair, confirmed by sketching, is correct.)

Double Integrals in Polar Coordinates

DP1 $x=r\cos\theta,\ y=r\sin\theta$. A polar "rectangle" $\Delta r\times\Delta\theta$ is a wedge whose arc length $r\,\Delta\theta$ grows with r, so its area is about $r\,\Delta r\,\Delta\theta$: equal steps sweep more area farther from the origin.

DP2 $\displaystyle\int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta=\int_0^{2\pi}\tfrac{1}{2}\,d\theta=\pi.$

DP3 $xy=\frac{r^2}{2}\sin 2\theta$, so $\int_0^{\pi/2}\int_0^2 \frac{r^3}{2}\sin 2\theta\,dr\,d\theta=\left(\int_0^2\frac{r^3}{2}dr\right)\left(\int_0^{\pi/2}\sin 2\theta\,d\theta\right)=2\cdot 1=2.$

DP4 In polar, $\sqrt{x^2+y^2} = r$ and a disk has constant limits $0 \le r \le a$, $0 \le \theta \le 2\pi$. In Cartesian, the square root of a sum does not simplify and the disk's boundary $y = \pm\sqrt{a^2 - x^2}$ makes both the integrand and the limits awkward.

DP5 Incorrect: it includes the inner disk $r < 1$, which is not in the annulus. Correct: $r : 1 \to 2$, $\theta : 0 \to 2\pi$.

DP6 $\displaystyle\int_{\pi/6}^{\pi/2}\int_0^3 f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta.$

Triple Integrals: Cylindrical and Spherical Coordinates

TI1 Cylindrical: $dV = r\, dr\, d\theta\, dz$ (clue: circular symmetry about an axis). Spherical: $dV = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta$ (clue: symmetry about a point, e.g. spheres or cones from the origin).

TI2 $\int_0^{2\pi}\int_0^2\int_0^5 r\, dz\, dr\, d\theta = 20\pi$ (matches $\pi r^2 h = \pi \cdot 4 \cdot 5$).

TI3 $\int_0^{2\pi}\int_0^{\pi/2}\int_0^1 \rho\cos\varphi\, \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta = 2\pi \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{\pi}{4}.$

TI4 On the cone $z = r$, i.e. $\rho\cos\varphi = \rho\sin\varphi$, so $\tan\varphi = 1$, $\varphi = \pi/4$. The solid is inside the cone: $0 \le \varphi \le \pi/4$, with $0 \le \rho \le \sqrt{2}$ and $0 \le \theta \le 2\pi$.

TI5 A cylinder has a constant circular cross-section along a straight axis, matching (r, θ, z) directly (r, θ fixed shape, z free). Spherical coordinates tie ρ and φ together in a way that does not respect the flat top and bottom, so φ-limits depend on ρ and the integrals become much messier.

TI6 E.g. a solid ball, or the shell between two concentric spheres: both are symmetric about the origin, so the limits are constants in ρ, φ, θ.

Module 3 — Vector Differentiation and Integration

Gradient

G1 $\nabla f = \dfrac{\partial f}{\partial x}\mathbf{i} + \dfrac{\partial f}{\partial y}\mathbf{j} + \dfrac{\partial f}{\partial z}\mathbf{k}$. It is perpendicular (normal) to the level surface $f = c$ at that point.

G2 $\nabla f = (3x^2y + z)\mathbf{i} + (x^3 - 2z^2)\mathbf{j} + (-4yz + x)\mathbf{k} = 5\mathbf{i} - \mathbf{j} + 9\mathbf{k}$.

G3 $\nabla f = (2x, 4y, 6z) = (2, 8, -6)$, so $2(x-1) + 8(y-2) - 6(z+1) = 0$, i.e. $x + 4y - 3z = 12$.

G4 Level curves are hyperbolas opening along the x-axis for $c > 0$, along the y-axis for $c < 0$, and a pair of crossing lines for $c = 0$. $\nabla f = (2x, -2y) = \mathbf{0}$ at the origin, a saddle point.

G5 False. Counterexample: $f = x^2 - y^2$ at $(0,0)$ has $\nabla f = \mathbf{0}$ but is a saddle point.

G6 E.g. $f = x^2 + 3y - z$, since $\nabla f = (2x, 3, -1) = (2, 3, -1)$ at $(1, 1, 1)$. Many answers are valid.

Directional Derivative

D1 $D_{\hat{\mathbf{a}}}f = \nabla f \cdot \hat{\mathbf{a}}$. It must have unit length so the result is a rate of change *per unit distance*, comparable across directions.

D2 $\nabla f = (y^2,\, 2xy + z,\, y) = (1, -1, -1)$; $\hat{\mathbf{a}} = (3, 4, 0)/5$. So $D = \frac{3}{5} - \frac{4}{5} = -\frac{1}{5}$.

D3 $\nabla f = (2xy^3, 3x^2y^2) = (4, 12)$. Maximum rate $= |\nabla f| = 4\sqrt{10}$, in the direction $(1, 3)/\sqrt{10}$.

D4 $\nabla f = (4, 2)$. $\cos\theta$ with $(1, 0)$ is $4/\sqrt{20} \approx 0.894$; with $(0, 1)$ it is $2/\sqrt{20} \approx 0.447$. $(1, 0)$ is closer in direction to ∇f, so it gives the larger value.

D5 $D_{\hat{\mathbf{a}}}f = \nabla f \cdot \hat{\mathbf{a}} = 0$ when $\hat{\mathbf{a}} \perp \nabla f$. At $(1, 1)$, $\nabla f = (2, 2)$; the unit vector $(1, -1)/\sqrt{2}$ gives $2/\sqrt{2} - 2/\sqrt{2} = 0$.

D6 E.g. $f = 3x + 4y$: $\nabla f = (3, 4)$ has magnitude 5 at every point.

Divergence

Dv1 $\operatorname{div}\mathbf{F} = \dfrac{\partial F_1}{\partial x} + \dfrac{\partial F_2}{\partial y} + \dfrac{\partial F_3}{\partial z}$. Positive: the point is a source (net outflow). Negative: a sink (net inflow).

Dv2 $2xy + 0 + 2yz = 2xy + 2yz$.

Dv3 $\operatorname{div}\mathbf{F} = 2 + 1 + \lambda = 0 \Rightarrow \lambda = -3$.

Dv4 $\operatorname{div}\mathbf{F} = 2$ everywhere. Divergence is a *local* measure at each point; for this linear field the spreading is uniform, not concentrated at the origin.

Dv5 It is solenoidal on the punctured region. But the Divergence Theorem requires $\mathbf{F}$ to be defined and differentiable throughout the enclosed volume; the singularity at the origin breaks this, so flux need not be zero despite $\operatorname{div}\mathbf{F} = 0$ elsewhere (the classic point-source case).

Dv6 E.g. $\mathbf{F} = -2x\mathbf{i} - 2y\mathbf{j} - 2z\mathbf{k}$.

Curl

C1 $\operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F}$, the 3×3 determinant with rows $(\mathbf{i}, \mathbf{j}, \mathbf{k})$, $(\partial_x, \partial_y, \partial_z)$, (F_1, F_2, F_3). Zero curl means no local rotation: a paddle wheel dropped anywhere would not spin.

C2 $\operatorname{curl}\mathbf{F} = (-2y - xy)\mathbf{i} + x\mathbf{j} + yz\mathbf{k}$; at $(1, 1, 1)$: $-3\mathbf{i} + \mathbf{j} + \mathbf{k}$.

C3 $\mathbf{i}$: $2 - 2 = 0$; $\mathbf{j}$: $-(2z - 2z) = 0$; $\mathbf{k}$: $2x - 2x = 0$. So $\operatorname{curl}\mathbf{F} = \mathbf{0}$: irrotational.

C4 Each component depends only on its own variable, so every cross-partial in the determinant (e.g. $\partial F_3/\partial y$) is 0, giving $\operatorname{curl}\mathbf{F} = \mathbf{0}$.

C5 Flawed. This is rigid-body-like rotation: nearby fluid elements genuinely spin about their own centres as they revolve, which is exactly what curl measures; the computed $2\mathbf{k}$ confirms it.

C6 E.g. $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$: $\operatorname{div}\mathbf{F} = 0$ but $\operatorname{curl}\mathbf{F} = 2\mathbf{k} \neq \mathbf{0}$.

Scalar and Vector Potential

P1 A scalar potential exists when $\operatorname{curl}\mathbf{F} = \mathbf{0}$ (irrotational); then $\mathbf{F} = \nabla\varphi$.

P2 Curl is $\mathbf{0}$. Integrating: $\varphi = x^2y + xz^3 + C$.

P3 Curl is $\mathbf{0}$. $\varphi = xy + yz + zx + C$.

P4 If $\nabla\varphi_1 = \nabla\varphi_2 = \mathbf{F}$ then $\nabla(\varphi_1 - \varphi_2) = \mathbf{0}$, so $\varphi_1 - \varphi_2$ is constant.

P5 Expected: $\operatorname{div}\mathbf{F} = 0$ does not imply $\operatorname{curl}\mathbf{F} = \mathbf{0}$, and a scalar potential needs the latter. For a solenoidal field, seek a *vector* potential $\mathbf{A}$ with $\mathbf{F} = \operatorname{curl}\mathbf{A}$.

P6 E.g. a constant field, or $\mathbf{F} = 2x\mathbf{i} - y\mathbf{j} - z\mathbf{k}$ ($\operatorname{div} = 2 - 1 - 1 = 0$; curl is $\mathbf{0}$ since each component depends only on its own variable).

Line Integrals

L1 $\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt$: the work done by $\mathbf{F}$ moving a particle along C.

L2 With $x = y = t$: $\mathbf{F}\cdot d\mathbf{r} = 2t\,dt$, so $\int_0^2 2t\,dt = 4$.

L3 With $x = t,\ y = t^3$: $\mathbf{F}\cdot d\mathbf{r} = 4t^3\,dt$, so the integral is 1.

L4 (a) -1; (b) $-\pi/2$. They differ because $\operatorname{curl}\mathbf{F} = -2\mathbf{k} \neq \mathbf{0}$: $\mathbf{F}$ is not conservative, so the integral is path-dependent.

L5 Yes. Reversing the direction of traversal flips the sign of $d\mathbf{r}$ at every point, so the integral changes sign and keeps its magnitude.

L6 Any non-conservative field works, e.g. $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ with the two paths from the previous analysis question.

Green's Theorem

Gr1 $\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA$, with C traversed counterclockwise around R.

Gr2 $\partial N/\partial x - \partial M/\partial y = 3x^2 - 3x^2 = 0$, so the integral is 0 (the field is conservative).

Gr3 Integrand $= (2xy + 2) - 2xy = 2$; area of the triangle $= \frac{1}{2}$. Answer: 1.

Gr4 M, N and their partials must be continuous throughout the enclosed region; this fails at the origin. Fix: cut out a small circle around the origin and apply the theorem on the resulting annular region.

Gr5 Green's Theorem: one double integral instead of six separate parametrised line integrals, especially when $\partial N/\partial x - \partial M/\partial y$ simplifies.

Gr6 E.g. $M = 0,\ N = x$ (or $M = -y/2,\ N = x/2$): $\partial N/\partial x - \partial M/\partial y = 1$.

Surfaces and Surface Integrals

S1 $dS = \sqrt{1 + g_x^2 + g_y^2}\, dA$. The root corrects for the surface's tilt: a tilted patch projects onto a smaller flat area.

S2 With $\hat{\mathbf{n}}\, dS = (1, 1, 1)\, dA$: $\mathbf{F} \cdot \hat{\mathbf{n}}\, dS = (2 - x - y)\, dA$, and $\int_0^1 \int_0^{1-x} (2 - x - y)\, dy\, dx = \dfrac{2}{3}$.

S3 $\int_0^{2\pi} \int_0^2 r\sqrt{1 + 4r^2}\, dr\, d\theta = \dfrac{\pi}{6}\left(17\sqrt{17} - 1\right) \approx 36.2$.

S4 $\mathbf{F} \cdot (-\hat{\mathbf{n}}) = -\mathbf{F} \cdot \hat{\mathbf{n}}$, so the value changes sign while $|\mathbf{F} \cdot \hat{\mathbf{n}}|$ is unchanged. The sign tells you whether net flow is into or out of a region, e.g. a control volume in fluid flow.

S5 A sphere has two z-values for each (x, y) in its shadow (upper and lower hemispheres). Fix: split into two graphs and treat each separately, or use a parametrisation such as spherical coordinates.

S6 E.g. S in the xy-plane ($\hat{\mathbf{n}} = \mathbf{k}$) and $\mathbf{F} = \mathbf{i}$: $\mathbf{F} \cdot \hat{\mathbf{n}} = 0$ everywhere.

Stokes' Theorem

St1 $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \hat{\mathbf{n}}\, dS$. Right-hand rule: curl the fingers along C, the thumb points along $\hat{\mathbf{n}}$.

St2 $\operatorname{curl} \mathbf{F} = (1, 1, 1)$, $\hat{\mathbf{n}} = (1, 1, 1)/\sqrt{3}$, so $(\operatorname{curl} \mathbf{F}) \cdot \hat{\mathbf{n}} = \sqrt{3}$. Triangle area $= \sqrt{3}/2$. Answer: $\dfrac{3}{2}$.

St3 $\operatorname{curl} \mathbf{F} = -z\mathbf{j} - y\mathbf{k}$; with $\hat{\mathbf{n}} = \mathbf{k}$ the integrand is $-y$. $\int_0^2 \int_0^2 (-y)\, dy\, dx = -4$.

St4 By Stokes' Theorem both equal the same line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ — the theorem does not care which surface bounded by C is used.

St5 It is 0: for any capping surface, $\iint_S (\operatorname{curl} \mathbf{F}) \cdot \hat{\mathbf{n}}\, dS = \iint_S \mathbf{0}\, dS = 0$, which is exactly the condition for $\mathbf{F}$ to be conservative.

St6 E.g. the unit circle in $z = 0$ with $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$ (as in the lesson notes): a flat disk has constant $\hat{\mathbf{n}} = \mathbf{k}$, avoiding a position-dependent normal vector.

Gauss Divergence Theorem

Gs1 $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS = \iiint_V \operatorname{div} \mathbf{F}\, dV$; S must be a *closed* surface enclosing the solid region V, with outward normal.

Gs2 Volume side: $\operatorname{div} \mathbf{F} = 6$, so 6. Surface side: faces $x = 1, y = 1, z = 1$ give $2, 1, 3$ and the opposite faces give 0: total 6. Equal. ✓

Gs3 $\operatorname{div} \mathbf{F} = 2x + 2y + 2z$, odd in each variable over a region symmetric about the origin, so the integral is 0.

Gs4 $\operatorname{div} \mathbf{F} = 0$ except at the origin. Applying the theorem to the region between the given surface and a small sphere about the origin shows the two fluxes are equal (no divergence

in between), so every enclosing surface carries the same flux as the small sphere.

Gs5 No: the theorem needs a closed surface. Add the flat circular base, compute the total flux via the theorem, then subtract the flux through the base to isolate the curved part.

Gs6 Any solenoidal field, e.g. a constant vector field $\mathbf{F} = \mathbf{c}$ ($\operatorname{div} \mathbf{c} = 0$).

Module 4 — Ordinary Differential Equations

Complementary Function (Homogeneous Case)

O1 Real distinct roots $m_1 \neq m_2$: $y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$. Repeated root m: $y = (C_1 + C_2 x)e^{mx}$. Complex roots $\alpha \pm i\beta$: $y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$.

O2 $m = 3, 4$: $y = C_1 e^{3x} + C_2 e^{4x}$.

O3 $(m + 3)^2 = 0$: $y = (C_1 + C_2 x)e^{-3x}$.

O4 $m = 2 \pm 3i$: $y = e^{2x}(C_1 \cos 3x + C_2 \sin 3x)$.

O5 (a) discriminant 0: repeated root, x-multiplied term. (b) discriminant < 0: complex roots, oscillatory. (c) discriminant > 0: distinct real roots, purely exponential.

O6 Mathematically equivalent (with complex constants), but the expected general *real* solution is $e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$: it needs only real constants and displays the decay and oscillation directly.

Undetermined Coefficients and Resonance

PI1 (a) e^{kx}: Ae^{kx}. (b) polynomial: a full degree-n polynomial. (c) $\sin kx, \cos kx$: $A \cos kx + B \sin kx$. Multiply by x (or x^2) whenever the guess already appears in the CF — resonance.

PI2 Try Ae^{3x}: $(9 - 9 + 2)A = 1$, so $y_{PI} = \frac{1}{2}e^{3x}$.

PI3 $y = C_1 \cos 2x + C_2 \sin 2x + 2x^2 - 1$.

PI4 CF $= (C_1 + C_2 x)e^{2x}$, so try $Ax^2 e^{2x}$: substitution gives $2A = 1$. $y = (C_1 + C_2 x)e^{2x} + \frac{1}{2}x^2 e^{2x}$.

PI5 $3e^{2x}$: $m = 2$ is a root, so resonance, try Axe^{2x}. $\sin x$: no root matches $\pm i$, so try $A \cos x + B \sin x$. The full PI is the sum.

PI6 The roots are $\pm i$, so $\sin x$ is already in the CF — resonance. Correct trial: $y_{PI} = x(A \cos x + B \sin x)$.

PI7 E.g. $y'' - 3y' + 2y = 0$, $y(0) = 2$, $y'(0) = 3$: $y = C_1 e^x + C_2 e^{2x}$ with $C_1 = C_2 = 1$, so $y = e^x + e^{2x}$.

Method of Variation of Parameters

VP1 With CF solutions y_1, y_2 and Wronskian $W = y_1y_2' - y_1'y_2$: $y_p = -y_1\int \frac{y_2R}{W}dx + y_2\int \frac{y_1R}{W}dx$. Preferred when $R(x)$ is not of the standard trial form ($\tan x,\ \sec x,\ 1/x,\ e^x/x, \dots$).

VP2 $y_1 = \cos x,\ y_2 = \sin x,\ W = 1$. $y_p = -\cos x\int \tan x\,dx + \sin x\int dx = \cos x \ln|\cos x| + x\sin x$. $y = C_1\cos x + C_2\sin x + \cos x\ln|\cos x| + x\sin x$.

VP3 $y_1 = e^x,\ y_2 = xe^x,\ W = e^{2x}$. $y_p = -e^x\int 1\,dx + xe^x\int \frac{1}{x}\,dx = -xe^x + xe^x\ln x$. The term $-xe^x$ is absorbed into the CF, so $y = (C_1 + C_2x)e^x + xe^x\ln x$.

VP4 $y_1 = \cos 2x,\ y_2 = \sin 2x,\ W = 2$. After integrating, $y_p = -\frac{1}{4}\cos 2x\ \ln|\sec 2x + \tan 2x|$ (the $\sin 2x\cos 2x$ terms cancel).

VP5 Repeated differentiation of $\tan x$ keeps producing new independent functions ($\sec^2 x,\ \sec^2 x\tan x, \dots$) so no finite trial family exists. Variation of parameters needs only the CF solutions and integrals, whatever $R(x)$ is.

VP6 Not wrong. A particular integral is not unique; $-xe^x$ is a CF term and is absorbed into C_2xe^x. The general solution is unchanged.

VP7 E.g. $y''+9y = \sec 3x$: $y_1 = \cos 3x,\ y_2 = \sin 3x,\ W = 3$, so $y_p = -\cos 3x\int \frac{\sin 3x\sec 3x}{3}dx + \sin 3x\int \frac{\cos 3x\sec 3x}{3}dx$.

Cauchy–Euler Equation

CE1 $x^2y'' + axy' + by = R(x)$. Put $x = e^t$ ($t = \ln x$), $D = \frac{d}{dt}$: $xy' = Dy$, $x^2y'' = D(D-1)y$. Homogeneous case: try $y = x^m$, giving $m(m-1) + am + b = 0$.

CE2 $m(m-1) - 2m - 4 = m^2 - 3m - 4 = 0$, $m = 4, -1$: $y = C_1x^4 + C_2x^{-1}$.

CE3 $m^2 + 2m + 1 = 0$: $m = -1$ repeated, $y = (C_1 + C_2\ln x)\dfrac{1}{x}$.

CE4 $m^2 + 1 = 0$, $m = \pm i$: $y = C_1\cos(\ln x) + C_2\sin(\ln x)$.

CE5 With $x = e^t$: $(D-2)^2y = e^{2t}$. CF $= (C_1 + C_2t)e^{2t}$; resonance gives PI $= \frac{1}{2}t^2e^{2t}$. So $y = x^2(C_1 + C_2\ln x) + \frac{1}{2}x^2(\ln x)^2$.

CE6 $t = \ln x \Rightarrow \dfrac{dy}{dx} = \dfrac{1}{x}\dfrac{dy}{dt}$, so $xy' = y_t$. Differentiating again: $y'' = -\dfrac{1}{x^2}y_t + \dfrac{1}{x^2}y_{tt}$, so $x^2y'' = y_{tt} - y_t$. Each power $x^ky^{(k)}$ becomes a constant-coefficient operator in t.

CE7 Wrong: the coefficients are variable, so e^{mx} gives no common factor to cancel. Use $y = x^m$: $m^2 - 4 = 0$, so $y = C_1x^2 + C_2x^{-2}$.

CE8 Need $m = 3, -2$: $m^2 - m - 6 = 0 = m(m-1) + am + b$ with $a = 0,\ b = -6$. So $x^2y'' - 6y = 0$.

Cauchy–Legendre Equation

CL1 $(ax+b)^2y'' + A(ax+b)y' + By = R(x)$. Put $ax + b = e^t$: $(ax+b)y' = a\,Dy$ and $(ax+b)^2y'' = a^2D(D-1)y$. Cauchy–Euler is the special case $a = 1,\ b = 0$.

CL2 Put $1+x=e^t$: $D(D-1)+D+1=D^2+1$, so $y=C_1\cos\ln(1+x)+C_2\sin\ln(1+x)$.

CL3 Put $2x+3=e^t$: $4(D^2-2D-3)y=3e^t-9$. CF $=C_1e^{3t}+C_2e^{-t}$; PI $=-\frac{3}{16}e^t+\frac{3}{4}$. So $y=C_1(2x+3)^3+C_2(2x+3)^{-1}-\frac{3}{8}x+\frac{3}{16}$.

CL4 Put $3x+2=e^t$: $9D(D-1)+9D-36=9(D^2-4)$, roots ±2. $y=C_1(3x+2)^2+C_2(3x+2)^{-2}$.

CL5 Wrong shift: the equation is in powers of $(x+1)$, so the trial is $y=(x+1)^m$: $m^2-1=0$, giving $y=C_1(x+1)+C_2(x+1)^{-1}$.

CL6 Need roots $2,-1$, i.e. D^2-D-2. With $2x+1=e^t$ (so $(2x+1)^2y''=4D(D-1)y$): $(2x+1)^2y''-8y=0$. Check: $y=(2x+1)^2$ has $y''=8$ and $8(2x+1)^2-8(2x+1)^2=0$.

Applications of Linear Second-Order ODEs

AP1 Spring–mass: $mx''+cx'+kx=F(t)$. LCR: $Lq''+Rq'+\dfrac{q}{C}=E(t)$. Regimes by c^2-4mk: >0 overdamped (two real roots), $=0$ critically damped (repeated root), <0 underdamped (oscillatory decay).

AP2 $x''+16x=0$, $\omega=4$: $x=0.3\cos4t$. Amplitude 0.3 m, period $2\pi/4=\pi/2$ s.

AP3 Repeated root -3: critically damped. $x=(C_1+C_2t)e^{-3t}$; $C_1=1$, $C_2-3=0$, so $x=(1+3t)e^{-3t}$.

AP4 $q''+4q'+5q=10$; roots $-2\pm i$, $q_p=2$. Initial conditions give $q=2-e^{-2t}(2\cos t+4\sin t)$. The charge tends to 2 C.

AP5 The driving frequency equals the natural frequency $\omega_0=2$ — resonance. $x_p=\frac{t}{4}\sin2t$, whose amplitude grows without bound (linearly in t).

AP6 Roots $-1\pm2i$ have a *negative real part*: $x=e^{-t}(\cdots)$ oscillates with decaying amplitude (underdamped). Oscillation forever occurs only with no damping ($c=0$).

AP7 Need $c^2=4mk=16$, so $c=4$: $x''+4x'+4x=0$, giving $x=(C_1+C_2t)e^{-2t}$.

Module 5 — Partial Differential Equations

Formation of Partial Differential Equations

FP1 Eliminating two arbitrary constants a,b from $z=F(x,y,a,b)$ (using p,q) gives a first-order PDE. Eliminating an arbitrary function from $\phi(u,v)=0$ gives a first-order linear (Lagrange) PDE $Pp+Qq=R$. Eliminating two arbitrary functions typically gives a second-order PDE.

FP2 $p=y+b$, $q=x+a$, so $z=pq$.

FP3 $p=a$, $q=b$, so $z=px+qy+p^2+q^2$.

FP4 $p=2xf'$, $q=2yf'$, so $p/q=x/y$: $yp-xq=0$.

FP5 Differentiating w.r.t. x and y and eliminating ϕ_u, ϕ_v: $(1+p)(y+zq)-(1+q)(x+zp)=0$, i.e. $(y-z)p+(z-x)q=x-y$.

FP6 $z_{xx}=f''+g''$ and $z_{yy}=f''+g''$, so $r=t$. Two arbitrary functions must be removed, and each needs one more differentiation than a single arbitrary function does, so the PDE is second order.

FP7 False. $z=-xy$ also satisfies the PDE ($p=-y,\ q=-x$: $px+qy+pq=-xy-xy+xy=-xy$) but is not of the two-parameter form for any constants a, b. It is the singular solution (the envelope of the family).

FP8 $p=-\dfrac{y}{x^2}f',\ q=\dfrac{1}{x}f'$, hence $xp+yq=0$.

Solutions of the Type $F(p,q)=0$

PQ1 Assume $z=ax+by+c$, so $p=a,\ q=b$. Then $F(a,b)=0$ lets you solve $b=\phi(a)$, giving the complete integral $z=ax+\phi(a)y+c$. The general integral comes from setting $c=\psi(a)$ and eliminating a via $\partial/\partial a=0$.

PQ2 $a+b=ab\Rightarrow b=\dfrac{a}{a-1}$. $z=ax+\dfrac{a}{a-1}\,y+c$.

PQ3 $\sqrt{b}=1-\sqrt{a}\Rightarrow b=(1-\sqrt{a})^2$ (valid for $0\le a\le 1$). $z=ax+(1-\sqrt{a})^2y+c$.

PQ4 $b=a^2$: $z=ax+a^2y+c$.

PQ5 The constants are tied by $F(a,b)=0$, which fixes b in terms of a; this leaves two free constants (a,c), matching the two independent variables.

PQ6 False. The cone $z=\sqrt{x^2+y^2}$ gives $p=x/r,\ q=y/r$, so $p^2+q^2=1$. It arises from the general integral (the envelope of the planes with $c=0$), not from any single plane.

PQ7 $p=a,\ q=1/a$, so $pq=1$.

Solutions of the Type $F(z,p,q)=0$

ZP1 Put $u=x+ay$ and $z=f(u)$. Then $p=\dfrac{dz}{du}$ and $q=a\dfrac{dz}{du}$, reducing the PDE to an ODE in z and u. Solve it and replace u by $x+ay$.

ZP2 $z'(1+az')=az'z\Rightarrow 1+az'=az$, so $\dfrac{dz}{az-1}=\dfrac{du}{a}$. Result: $az-1=Ce^{x+ay}$, i.e. $z=\dfrac{1+Ce^{x+ay}}{a}$.

ZP3 $z^2\left[(1+a^2)z'^2+1\right]=1\Rightarrow\dfrac{z\,dz}{\sqrt{1-z^2}}=\dfrac{du}{\sqrt{1+a^2}}$. Result: $(x+ay+b)^2=(1+a^2)(1-z^2)$.

ZP4 $az'^2=z\Rightarrow\dfrac{dz}{\sqrt{z}}=\dfrac{du}{\sqrt{a}}$, so $2\sqrt{z}=\dfrac{u+b}{\sqrt{a}}$. Result: $4az=(x+ay+b)^2$.

ZP5 With no explicit x, y, the equation is unchanged by translations in the (x,y)-plane, so solutions can be sought that depend on x, y only through one travelling combination $x+ay$. An explicit x or y would spoil this invariance.

ZP6 Invalid: the equation contains x explicitly, so it is not of type $F(z,p,q)=0$ and the substitution does not reduce it to an ODE in u alone.

ZP7 E.g. $p^2+q^2=z^2$: $(1+a^2)z'^2=z^2$, so $z=C\exp\Big(\dfrac{x+ay}{\sqrt{1+a^2}}\Big)$.

Solutions of the Type $F(x,p)=G(y,q)$

XP1 Set both sides equal to a constant a: $F(x,p)=a=G(y,q)$. Solve $p=f(x,a)$ and $q=g(y,a)$, then integrate $dz=p\,dx+q\,dy$ to get $z=\int f\,dx+\int g\,dy+b$.

XP2 $p^2-x=y-q^2=a$: $p=\sqrt{x+a},\ q=\sqrt{y-a}$. $z=\frac{2}{3}(x+a)^{3/2}+\frac{2}{3}(y-a)^{3/2}+b$.

XP3 $\dfrac{p^2(1+x^2)}{x^2}=\dfrac{q}{y}=a$: $p=\dfrac{\sqrt{a}\,x}{\sqrt{1+x^2}},\ q=ay$. $z=\sqrt{a}\sqrt{1+x^2}+\dfrac{ay^2}{2}+b$.

XP4 $pe^{-x}=qe^{-y}=a$: $p=ae^x,\ q=ae^y$. $z=a(e^x+e^y)+b$.

XP5 The left side involves only x (and p), the right only y (and q). Seeking solutions where p depends on x alone and q on y alone, a function of x equal to a function of y for all x,y must be constant.

XP6 Wrong: rearrange to $p-x=y-q$, which is of the form $F(x,p)=G(y,q)$. Then $p=x+a,\ q=y-a$ and $z=\frac{x^2}{2}+ax+\frac{y^2}{2}-ay+b$.

XP7 $p=2(x+a),\ q=2(y-a)$, so $p-2x=2a=2y-q$, i.e. $p+q=2(x+y)$.

Clairaut's Form and the Singular Solution

CK1 $z=px+qy+f(p,q)$. Complete integral: replace p,q by constants: $z=ax+by+f(a,b)$. The singular solution is found by differentiating w.r.t. a and b and eliminating a,b.

CK2 Complete: $z=ax+by+a^2+b^2$. Differentiate: $x+2a=0,\ y+2b=0$, so $a=-x/2,\ b=-y/2$ and $z=-\dfrac{x^2+y^2}{4}$, i.e. $4z=-(x^2+y^2)$.

CK3 Complete: $z=ax+by+\sqrt{1+a^2+b^2}$. With $s=\sqrt{1+a^2+b^2}$: $x=-a/s,\ y=-b/s$, $z=1/s$, so $x^2+y^2+z^2=1$ (the unit sphere).

CK4 $z=px+qy+\dfrac{1}{p+q}$. Complete integral: $z=ax+by+\dfrac{1}{a+b}$.

CK5 It is the *envelope* of the family: a paraboloid that is tangent to each plane of the family. It is not obtainable from the complete integral by choosing constants.

CK6 Invalid: a,b must be constants, not functions of y,x. $z=-xy$ is the singular solution (envelope), a genuinely different solution that is not in the family.

CK7 Tangent planes: $z=ax+by-\dfrac{a^2+b^2}{4}$ (with $a=2x_0,\ b=2y_0$). The PDE is $z=px+qy-\dfrac{p^2+q^2}{4}$.

Lagrange's Linear Equation $Pp+Qq=R$

LG1 $Pp+Qq=R$. Auxiliary equations: $\dfrac{dx}{P}=\dfrac{dy}{Q}=\dfrac{dz}{R}$. If $u(x,y,z)=c_1$ and $v(x,y,z)=c_2$ are two independent solutions, the general solution is $\phi(u,v)=0$ (or $v=f(u)$).

LG2 $\dfrac{dx}{x}=\dfrac{dy}{y}=\dfrac{dz}{z}$ gives $\dfrac{y}{x}=c_1,\ \dfrac{z}{x}=c_2$. General solution: $\phi\big(\frac{y}{x},\frac{z}{x}\big)=0$, i.e. $z=x\,f(y/x)$.

LG3 $\dfrac{dx}{yz}=\dfrac{dy}{zx}=\dfrac{dz}{xy}\Rightarrow x\,dx=y\,dy=z\,dz$. So $x^2-y^2=c_1,\ y^2-z^2=c_2$: $\phi(x^2-y^2,\ y^2-z^2)=0$.

LG4 Multipliers $1,1,1$: $d(x+y+z)=0$, so $x+y+z=c_1$. Multipliers x,y,z: $x\,dx+y\,dy+z\,dz=0$, so $x^2+y^2+z^2=c_2$. Solution: $\phi(x+y+z,\ x^2+y^2+z^2)=0$ (the same PDE as the formation question).

LG5 From $\dfrac{d(x-y)}{-(x-y)}=\dfrac{d(y-z)}{-(y-z)}$: $\dfrac{x-y}{y-z}=c_1$. Using $\dfrac{d(x+y+z)}{2(x+y+z)}=-\dfrac{d(x-y)}{x-y}$: $(x-y)^2(x+y+z)=c_2$. Solution: $\phi\Big(\dfrac{x-y}{y-z},\ (x-y)^2(x+y+z)\Big)=0$.

LG6 A surface $z=z(x,y)$ has normal $(p,q,-1)$, and $Pp+Qq=R$ says $(P,Q,R)\perp(p,q,-1)$, i.e. the vector (P,Q,R) is tangent to the surface. The auxiliary equations describe curves (characteristics) tangent to (P,Q,R); $u=c_1,\ v=c_2$ is such a curve. A surface built from a one-parameter family of these curves, $\phi(u,v)=0$, is therefore an integral surface.

LG7 Incorrect. The general solution is $\phi(u,v)=0$ with ϕ an *arbitrary* function; $u+v=c$ is just one particular member, giving only a one-parameter family of solutions.

LG8 $yp-xq=0$ $(P=y,\ Q=-x,\ R=0)$: $dz=0\Rightarrow z=c_1$, and $x\,dx+y\,dy=0\Rightarrow x^2+y^2=c_2$.

Linear First-Order PDEs by Separation of Variables

SV1 Assume $u=X(x)Y(y)$ (or $X(x)T(t)$), substitute, and rearrange so each side depends on one variable only; set both equal to a separation constant k. Solve the two ODEs, form $u=XY$, and fix constants from the given condition.

SV2 $\dfrac{X'}{X}=2\dfrac{T'}{T}+1=k$: $X=e^{kx},\ T=e^{(k-1)t/2}$. $u(x,0)=Ce^{kx}=6e^{-3x}\Rightarrow C=6,\ k=-3$. So $u=6e^{-3x-2t}$.

SV3 $\dfrac{X'}{X}=-\dfrac{2}{3}\dfrac{Y'}{Y}=k$: $u=Ce^{kx-3ky/2}$. The condition gives $C=4,\ k=-1$. So $u=4e^{-x+3y/2}$.

SV4 $\dfrac{4X'}{X}=3-\dfrac{Y'}{Y}=k$: $u=Ce^{kx/4+(3-k)y}$. $u(0,y)=Ce^{(3-k)y}=e^{-5y}\Rightarrow C=1,\ k=8$. So $u=e^{2x-5y}$.

SV5 The PDE is linear, so solutions superpose. Each term matches its own k: $k=-3$ gives e^{-3x-2t}, and $k=-5$ gives e^{-5x-3t}. So $u=6e^{-3x-2t}+4e^{-5x-3t}$.

SV6 Incorrect. A function of x equal to a function of t must be a *constant* k, not necessarily zero. The value of k is then fixed by the given condition.

SV7 E.g. $3u_x - 2u_y = 0$ with $u(x,0) = 5e^{2x}$: $u_x = 2u,\ u_y = 3u$, so $6u - 6u = 0$.